

Exploring Exponential Simulations and Their Center and Spread

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2025-04-05

Overview

This document explores the exponential distribution through simulations in R. We generate both individual exponential values and averages from multiple samples to compare their properties with theoretical expectations. The study uses histograms and overlaid density curves to visualize the differences between raw exponential data and the normally distributed sample means predicted by the Central Limit Theorem.

Distributions

We will create two simulations to investigate and compare their shape, mean, and variance.

First Distribution - Single Sample of Exponential Variables

The first distribution is 1000 random exponential variables with $\lambda = 0.2$. Given our knowledge of exponential functions, we expect that when we visualize this distribution it will resemble the theoretical density function $f(x) = 0.2e^{-0.2x}$.

First we generate the variables.

```
#set lambda value
lambda <- 0.2
#simulation of 1000 random exponential values
exp_raw <- rexp(1000, rate = lambda)
```

The first few entries of our simulation: 4.2172863, 2.8830514, 6.6452743, 0.1578868, 0.2810549, 1.5825061

We can see that the values appear to be drawn from an exponential distribution.

Now, let's calculate the theoretical mean and variance for an exponential distribution with $\lambda = 0.2$.

```
#calculate theoretical mean
theoretical_mean <- 1/lambda
#calculate theoretical variance
theoretical_sd <- 1/lambda
theoretical_var <- (1/lambda)^2
```

The Theoretical Mean: 5

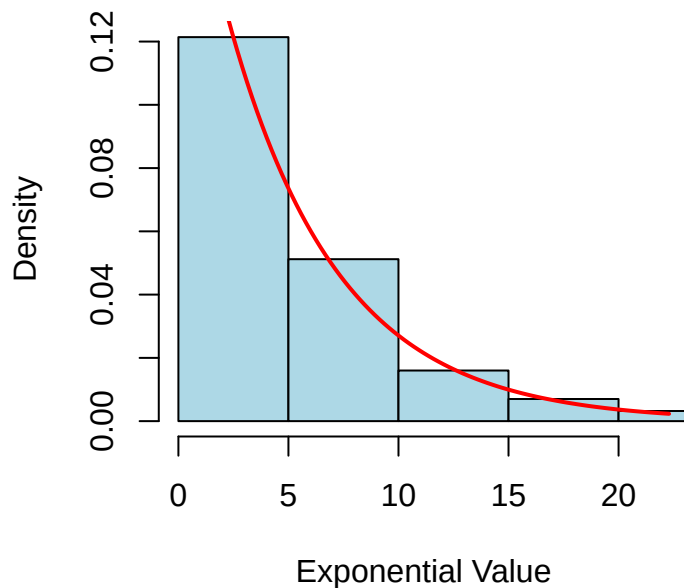
The Theoretical Variance: 25

Next, we view the distribution to see if it reflects our expectation of being right-skewed and decreasing, similar in shape to the theoretical exponential function. To facilitate this, I overlay the function $f(x) = 0.2e^{-0.2x}$ on the histogram.

```
#histogram form 1000 exponential values
hist(exp_raw, probability = TRUE,
      main = "Histogram of 1000 Exponential Values",
      xlab = "Exponential Value",
```

```
col = "lightblue",
xlim = c(0, quantile(exp_raw, 0.99))) #create a common scale for the function
#overlay the theoretical exponential density curve
curve(dexp(x, rate = lambda),
      from = 0,
      to = quantile(exp_raw, 0.99), #apply the common scale to the function
      add = TRUE,
      col = "red",
      lwd = 2)
```

Histogram of 1000 Exponential Values



This density function is right-skewed and decreasing, and the overlay shows a close fit to the theoretical distribution.

The Law of Large Numbers (LLN) tells us that if the sample is large enough, the sample distribution will approach the theoretical density curve. We can see that our sample approximates the theoretical curve well, indicating that $n=1000$ is a good sample size.

To further support this, let's compare the theoretical mean and variance to the sample mean and variance.

Sample Mean: 5.1498964

Sample Variance: 25.2207028

Again, we observe that the sample and theoretical values are very similar.

Second Distribution - 1000 Sample Means of 40 Exponential Variables

This second simulation consists of 1000 samples, each containing 40 exponential random variables with $\lambda = 0.2$. We then compute the mean for each sample (each column of a matrix) to create a distribution of 1000 sample means. First, let's generate the values.

```
#generate 1000 simulation of 40 random exponential variables in a matrix where each column contains 40
n <- 40
n_sim <- 1000
sim_matrix <- matrix(rexp(n*n_sim, rate = lambda), nrow = n, ncol = n_sim)
#calculate the mean of each sample by finding the average of each column (this is our new data set)
```

```
sim_means <- apply(sim_matrix, 2, mean)
#calculate the variance of each sample (just for fun)
sim_vars <- apply(sim_matrix, 2, var)
```

First few averages from each sample: 4.8019961, 4.4261485, 4.4172168, 4.967223, 5.4704708, 5.6579778

First few variances from each sample: 20.8808216, 24.3205689, 19.2878375, 21.5869698, 19.2398573, 47.8443771

We observe that, as expected, the sample means are approximately 5 and the sample variances are about 25.

We will only use the sample averages moving forward.

Now, let's compare the sample mean and variance of these averages to their theoretical counterparts.

The mean of the averages will be $1/\lambda$ as before. However the variance is the standard error squared or $\left(\frac{1}{\lambda\sqrt{n}}\right)^2$ because this distribution is the averages of 1000 sample means.

```
#theoretical mean calculation
lambda <- 0.2
theoretical_mean <- 1/lambda
#theoretical variance calculation
n <- 40
theoretical_sd <- 1/lambda
theoretical_se <- theoretical_sd/sqrt(n)
theoretical_mean_var <- theoretical_se^2
```

Theoretical Mean: 5

Theoretical Variance of the Sample Mean: 0.625

Now let's compare the sample means and variance to the theoretical values

```
sample_mean <- round(mean(sim_means), 3)
sample_var <- round(var(sim_means), 3)
```

Sample Mean: 5.013

Sample Variance: 0.604

As you can see, the sample mean and variance are extremely close to the theoretical values. This suggests that the distribution of sample means is approximately normal.

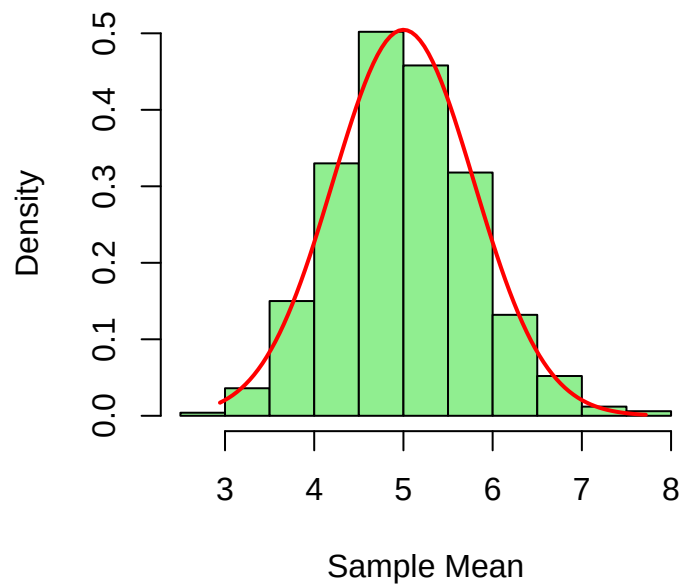
Lastly, let's visualize the distribution of the sample means. Unlike the first simulation, which generated 1000 random exponential values, this simulation generated 1000 sample means.

According to the Central Limit Theorem (CLT), even though individual exponential observations are not normally distributed, the distribution of the sample mean will tend to a normal distribution as the sample size increases. With 1000 samples of 40 observations each, we expect a nearly normal distribution.

For comparison, I overlay the theoretical normal distribution (with the theoretical mean and variance) on the histogram of the sample means.

```
#plot the histogram of sample means with density scaling
hist(sim_means, probability = TRUE,
     main = "Histogram of 1000 Sample Means (n = 40)",
     xlab = "Sample Mean",
     col = "lightgreen")
#overlay the theoretical normal density curve
curve(dnorm(x, mean = theoretical_mean, sd = sqrt(theoretical_mean_var)),
     from = min(sim_means), to = max(sim_means),
     add = TRUE, col = "red", lwd = 2)
```

Histogram of 1000 Sample Means ($n = 40$)



We can see that the sample distribution fits the theoretical model well and is normally distributed as expected.

Summary

In conclusion, our simulations confirm that while individual exponential data are right-skewed, the distribution of sample means tends toward normality. The sample means and variances closely match the theoretical values, demonstrating the accuracy of our simulation.